ACADEMIC TERM PROJECT • BIG DATA SYSTEMS & DISTRIBUTED COMPUTING

Project Initiation Document (PID)

AutoData Lake: Distributed Object Storage, Consistent Hashing & Telemetry Lakehouse Architecture

EXECUTIVE SCOPE: This project establishes a resilient, distributed Big Data Lakehouse engineered to handle the high velocity, massive volume, and multimodal variety of connected vehicle telematics. Instead of relying on monolithic relational databases or superficial API wrappers, this architecture implements foundational distributed algorithms from first principles—including **Consistent Hashing with Virtual Nodes** for dynamic partition re-sharding, **Content-Addressable Storage (CAS via SHA-256)** for multi-gigabyte deduplication, **Scalable In-Memory Bloom Filters** for zero-I/O fault verification, and **Columnar Storage (Apache Parquet)** for high-performance distributed analytics.

Course & Degree:	Big Data Systems (CS/DS) — Bachelor of Science in Data Science (BSDS)
Department / Faculty:	Department of Data Science / Computing
Project Domain:	Distributed Systems, Object Storage, Stream Ingestion & Lakehouse Analytics
Document Classification:	Formal Term Project Initiation Document (PID) • Fall 2026

Project Team & Assigned Engineering Responsibilities

Sr.	Member Name	Roll No.	Primary Assigned Technical Responsibilities
1	Talha Ali Bhatti	**bsdsf23a016**	Distributed Storage & Streaming Lead: MinIO S3 Object Store cluster configuration, bucket lifecycle policies, Kafka/Stream ingestion buffer, and Parquet partitioning.
2	Muhammad Abdullah	**bsdsf23a032**	Data Engineering & Analytics Lead: Distributed batch ETL pipelines (PySpark/DuckDB), fleet-wide predictive anomaly models, and OLAP query benchmarking.
3	Abdullah	**bsdsf23a035**	Distributed Algorithms Lead: Consistent Hashing Ring engine (Virtual Nodes/Ketama), Content-Addressable Storage (CAS SHA-256) deduplication, and Bloom Filter index.

1. The 5 V's of Big Data in Connected Telematics

Dimension	Workload Manifestation in Connected Vehicles	Architectural Solution in AutoData Lake
Volume	10,000+ vehicles generating ~50MB daily telemetry + 20GB unstructured OEM service manual PDFs & damage photos.	Distributed Object Storage (MinIO S3) decoupled from compute; storage compression via Apache Parquet.
Velocity	Continuous CAN-bus & OBD-II sensor streams arriving at 10 Hz to 50 Hz per vehicle requiring sub-second safety triage.	Asynchronous Dual-Speed Ingestion (Speed Buffer) utilizing sliding-window micro-batching.
Variety	Structured telemetry (Parquet), Unstructured binary media (PDF manuals, cluster photos), and Semi-structured logs (DTCs).	Unified Multi-Modal Lakehouse storing structured time-series and binary blobs with unified catalogs.
Veracity	Sensor dropouts, noise, corrupt frames, and redundant manual uploads across owners.	SHA-256 cryptographic verification for Content-Addressable Storage (CAS) and schema enforcement.
Value	Driver safety triage, fleet predictive failure warning, and localized Pakistani PKR repair cost benchmarks.	Distributed OLAP analytics (DuckDB/PySpark) driving component longevity models & workshop price protection.

2. Foundational Distributed Systems Concepts Implemented

2.1 Distributed Object Storage (MinIO S3 API): Replaces POSIX file systems with cloud-native S3 object store for manuals, images, and partitioned telemetry.

2.2 Content-Addressable Storage (CAS SHA-256): Files stored by SHA-256 hash. Duplicate manual uploads share single physical block, reducing unstructured disk overhead by 40%–60%.

2.3 Consistent Hashing Ring (Ketama with Virtual Nodes): Vehicle VINs mapped across [0, 2^32 - 1] ring with V=150 virtual nodes per instance. Migrates only K/N keys upon node addition/eviction.

2.4 Scalable In-Memory Bloom Filter: Pre-indexes 10,000+ SAE diagnostic trouble codes (DTCs) with zero false negatives and $p < 0.01$, intercepting non-existent queries before storage I/O.

2.5 Columnar Lakehouse Storage (Apache Parquet + Snappy): Encodes telemetry with dictionary compression and run-length encoding, delivering >70% storage reduction and fast column projection.

3. RACI Responsibility Assignment Matrix

Technical Task / Deliverable	Talha Ali Bhatti (016)	M. Abdullah (032)	Abdullah (035)
Distributed Object Store Configuration (MinIO S3)	**Accountable (A)**	Informed (I)	Consulted (C)
Consistent Hashing Ring & Virtual Nodes Engine	Consulted (C)	Informed (I)	**Accountable (A)**
Content-Addressable Storage (CAS SHA-256) Engine	Informed (I)	Consulted (C)	**Accountable (A)**
Scalable In-Memory Bloom Filter Lookup Engine	Informed (I)	Consulted (C)	**Accountable (A)**
High-Frequency Stream Ingestion Buffer & Telemetry	**Accountable (A)**	Consulted (C)	Informed (I)
Apache Parquet Columnar Lakehouse & Partitioning	**Accountable (A)**	Consulted (C)	Consulted (C)
Distributed Fleet Analytics Pipeline (PySpark/DuckDB)	Consulted (C)	**Accountable (A)**	Informed (I)
Empirical Benchmarking & Chi-Square Analysis	Consulted (C)	**Accountable (A)**	Consulted (C)
Final Documentation, Scientific Report & Slides	Responsible (R)	Responsible (R)	Responsible (R)

Submitted on behalf of Group BSDS-23-AutoData: Talha Ali Bhatti (bsdsf23a016) • Muhammad Abdullah (bsdsf23a032) • Abdullah (bsdsf23a035)